

Md. Shajibul Islam

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RESEARCH INTERESTS

Computer Vision; Edge AI; Applied Deep Learning; Intelligent Transportation Systems; Natural Language Processing; AI for Healthcare

EDUCATION

East West University, Dhaka, Bangladesh Jan. 2019 - Sept. 2023
B.Sc. in Computer Science and Engineering, Major: Intelligent Systems and Data Science | GPA: 3.60/4.00

RESEARCH EXPERIENCE

DhakaRoadNet - Edge AI Road-Object Detection May 2026 - June 2026

- Created and verified a Dhaka road-scene dataset containing 2,145 images, 23,448 annotated objects, and 24 traffic-related classes.
- Trained and evaluated YOLOv8n (validation mAP50: 0.798; test mAP50: 0.743), analyzed weak classes, and exported an FP16 TensorFlow Lite model.
- Deployed fully offline image, camera, and live-video inference in a native Android application; first author of the submitted manuscript.

Undergraduate Thesis Research, East West University Sept. 2022 - Sept. 2023
Thesis: "Image Processing & Deep Learning Based Road Object Detection System for Safe Transportation"

Supervisor: Dr. Md. Nawab Yousuf Ali

- Investigated image-processing and deep-learning methods for detecting road objects in support of safer transportation; the work was subsequently published at ICCCNT 2024.

PUBLICATIONS AND MANUSCRIPT

- "DhakaRoadNet: An Offline Edge AI Framework for Road-Object Detection in Dhaka Traffic Using YOLOv8n and TensorFlow Lite." Manuscript submitted to COMPAS 26; First Author.
- R. Gani et al., "Impact of AI in Healthcare Services: Analysis Using Medical Synthetic Data," in *2024 International Conference on Advances in Computing, Communication, Electrical, and Smart Systems (iCACCESS)*, Dhaka, Bangladesh, 2024, pp. 1-6. doi:10.1109/iCACCESS61735.2024.10499613.
- R. H. Prito, M. S. Hossain, **M. S. Islam**, M. Meem, and M. N. Y. Ali, "Image Processing and Deep Learning Based Road Object Detection System for Safe Transportation," in *2024 15th International Conference on Computing Communication and Networking Technologies (ICCCNT)*, 2024, pp. 1-7. doi:10.1109/ICCCNT61001.2024.10726014.
- R. H. Prito, M. S. Hossain, **M. S. Islam**, M. Meem, and A. W. Reza, "Liver Tumor Detection Using CLAHE Histogram Equalization and U-Net Model from 3D CT Scan Images," in *Intelligent Computing and Optimization, Lecture Notes in Networks and Systems*, vol. 1169, Springer, 2024. doi:10.1007/978-3-031-73324-6_9.
- M. R. A. Rashid, M. S. Hossain, M. Fahim, **M. S. Islam**, T.-E. Alindo, R. H. Prito, M. S. A. Sheikh, M. S. Ali, M. Hasan, and M. Islam, "Comprehensive Dataset of Annotated Rice Panicle Image from Bangladesh," *Data in Brief*, vol. 51, art. 109772, 2023. doi:10.1016/j.dib.2023.109772.

TEACHING EXPERIENCE

Graduate Teaching Assistant, East West University Jan. 2023 - Dec. 2023

Courses: Data Science, Artificial Intelligence, and Machine Learning

- Supported laboratory instruction, coding assignments, debugging, and practical demonstrations for undergraduate and graduate students.
- Mentored semester projects and research papers; reviewed academic writing, citations, conference submissions, and journal manuscripts.

SELECTED RESEARCH PROJECTS

ScholarRAG - Local Citation-Grounded Research Paper Q&A [GitHub] Apr. 2026 - May 2026

- Built a fully local RAG pipeline for English research papers using page-aware extraction and chunking, sentence-transformer embeddings, ChromaDB retrieval, and an Ollama-hosted language model.
- Generated evidence-grounded answers with paper/page citations and implemented retrieval evaluation using Page Hit@K, hit rate, and mean reciprocal rank.

Ames Housing Regression Benchmark [GitHub] Jan. 2026 - Apr. 2026

- Developed a reproducible preprocessing, feature-engineering, and model-comparison pipeline; stacking achieved the best reported performance (R^2 0.8877, RMSE 0.1448, MAE 0.0939).

PROFESSIONAL EXPERIENCE

Pathao Limited, Dhaka, Bangladesh

Jan. 2024 - Present

Software Engineer I (Jan. 2025 - Present); **Associate Software Engineer** (May 2024 - Dec. 2024); **Software Engineer Intern** (Jan. 2024 - Apr. 2024)

- Develop and maintain production Android systems in Kotlin/Java, integrating REST services, local persistence, concurrency, and modular application architectures.
- Contributed to driver onboarding, low-connectivity ride workflows, location-based services, restaurant operations, and internal SDK reliability in collaboration with engineering, product, and design teams.

PROGRAMMING COMPETITIONS AND HONORS

- Solved 1,500+ problems across Codeforces, LeetCode, and HackerRank; participated in 200+ online and 20+ onsite contests.
- 1st Runner-Up, EWU In-House Programming Battle (2021); 2nd Runner-Up, EWU Victory Day Programming Battle (2021), EWU Intra-University Contest (2022), and EWU Intra Kick-Off Contest (2022).
- Honorable Mention, BUET Inter-University Programming Contest (2022); Top 15%, ICPC Asia Dhaka Regional Online Preliminary Contest (2022).

TECHNICAL SKILLS

Languages: Python, Kotlin, Java, C++, SQL

ML / AI: PyTorch, TensorFlow Lite, scikit-learn, YOLO, OpenCV, sentence transformers; computer vision, object detection, NLP, RAG, model evaluation

Systems and Tools: Android SDK, Git/GitHub, Jupyter, Google Colab, LaTeX, ChromaDB, Ollama